MACHUM ROY CHOUDHURY

Vellore, Tamil Nadu, India | machumroy.choudhury2023@vitstudent.ac.in
linkedin.com/in/machum-roy-choudhury-463999270 | github.com/MRC005

PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science and Engineering student at VIT Vellore (VITEEE 2023 All India Rank 413) with a government software development internship at AMTRON and six end-to-end projects across full-stack web development (MERN), Java and cloud systems, machine learning, computer vision, NLP, and embedded IoT. Proficient in Java, Python, C++, and JavaScript with a strong Data Structures and Algorithms foundation and full software development lifecycle exposure. Seeking a Software Engineer role building scalable, production-grade systems.

EDUCATION

Vellore Institute of Technology (VIT), Vellore, Tamil Nadu **2023 – 2027 (Expected)**
B.Tech, Computer Science and Engineering (Core) | VITEEE 2023: All India Rank 413

TECHNICAL SKILLS

Programming Languages: Java, Python, C++, C, JavaScript, SQL
Web & Full-Stack: React.js, Node.js, Express.js, MongoDB, MERN Stack, REST APIs, HTML5, CSS3, JSON, Responsive UI Design
Machine Learning & AI: Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Feature Engineering, SVM, Computer Vision, NLP, Model Evaluation
Cloud, Data & DevOps: Data Engineering, Data Pipelines, Telemetry Processing, CloudSim Plus, Cloud Cost Optimisation, Git, GitHub, Version Control, Maven, Build Automation
Embedded & IoT: Arduino, Embedded C++, PID Control, DS18B20 Sensors, MOSFET Drivers, I2C, Real-Time Control Systems
Core CS & Tools: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design, SDLC, Debugging & Testing, Postman, VS Code, MATLAB, Linux
Soft Skills: Problem Solving, Critical Thinking, Analytical Reasoning, Team Collaboration, Stakeholder Communication, Technical Documentation, Adaptability, Ownership, Time Management, Leadership

INTERNSHIP EXPERIENCE

Software Development Intern — AMTRON (Assam Electronics Development Corporation Ltd.) **2026**
Guwahati, Assam | Government of Assam undertaking, Dept. of Information Technology — nodal ICT and e-governance implementing agency for North East India

- Contributed to **live ICT and e-governance software applications** serving multiple Government of Assam departments as part of an established engineering team.
- Executed the **full software development lifecycle** — requirements gathering, implementation, testing, debugging, and delivery — while communicating technical progress to project stakeholders.
- Gained practical exposure to **large-scale, government-grade system architecture** and institutional software delivery workflows.

PROJECTS

MinuteMeals — Full-Stack Food Delivery Platform
Tech Stack: *React.js, Node.js, Express.js, MongoDB, JavaScript, REST APIs | github.com/MRC005/MinuteMeals*

- Engineered an **end-to-end MERN web application** with real-time order tracking and a RESTful API backend, owning the full stack solo from database schema to deployment.
- Built an **admin dashboard** for menu management, order-lifecycle control, and operational monitoring, alongside a responsive, mobile-first React.js interface.

Autonomous Cloud Cost Governor Agent
Tech Stack: *Java (JDK 17+), CloudSim Plus, Maven, Python, Pandas, Matplotlib | github.com/MRC005/cloudsim-autonomous-cost-governor*

- Simulated a multi-VM **cloud datacenter in Java** using CloudSim Plus, instrumenting CPU, RAM, and bandwidth cost telemetry into structured CSV data pipelines.
- Developed a **Cost Failure Probability predictor** from utilisation signals plus an autonomous agent visualising CPU cost and idle-waste trends, proving predictive over reactive cost management.

Tyre Tread Wear Classification — Computer Vision & Machine Learning
Tech Stack: *Python, OpenCV, Scikit-learn, GLCM, TSCI, SVM, Edge Detection | github.com/MRC005/tyre-tread-analysis*

- Achieved **74.8% classification accuracy** separating worn from serviceable tyres on a 369-image dataset using 5-fold stratified cross-validation.
- Engineered a **three-stage feature-fusion pipeline** (TSCI frequency, GLCM texture, edge density); ablation confirmed fusion beat GLCM+Edge (72.6%) and single-feature baselines (63–66%).

Real-Time Precision Cooling System — Arduino PID Controller
Tech Stack: *Embedded C++, PID Control, DS18B20, I2C LCD, MOSFET Drivers | github.com/MRC005/Real-Time-Precision-Cooling-System*

- Designed a **four-mode closed-loop temperature controller** (Manual, Linear, PID, AUTO) with tuned PID gains (Kp=8.0, Ki=0.3, Kd=2.0), anti-windup clamping, a hybrid AUTO mode switching strategy on error thresholds, and a rotary-encoder UI on a 16×2 I2C LCD.

Additional Projects: Cross-Lingual Semantic Role Labeling via Zero-Shot Transfer (NLP, multilingual transfer learning) | **DomoticsHub — IoT Smart Home Automation** (Python, React.js, IoT protocols).

ACHIEVEMENTS, LEADERSHIP & CERTIFICATIONS

- **VITEEE 2023 — All India Rank 413** in a national engineering entrance examination; **shortlisted for interview at TECDIA India** via a competitive coding challenge evaluation.
- **Core Member, VIT** — Dream Merchants (finance and external partnerships), AdMark (creative direction and publicity), and Enajori Assamese Literary Association (event management), leading cross-functional student teams.
- **Certifications:** MERN Stack Web Development (FacePrep, 2025); AutoML Bootcamp (Udemy, 2023); Cyber Security Workshop (VIT graVITas, 2023).
- **Languages:** English and Hindi (Professional), Assamese (Native), Bengali (Limited Working), Japanese (Elementary).